

# E867981\_92

The equation  $2x^4 + x + y^3 - y^2 - 2 = 0$  has no integer solutions

Size  $h = 48$ , length  $l = 12$

**Theorem 1.** *The equation*

$$2x^4 + x + y^3 - y^2 - 2 = 0 \tag{1}$$

*has no integer solutions.*

The proof uses a polynomial identity of the form  $HT = U^2 + 15V^2$  on the curve (1), and the fact that an odd prime  $p$  with  $(-15/p) = -1$  divides  $U^2 + 15V^2$  only to an even power. Apart from polynomial identities, which can be checked by expanding both sides, and finite checks of congruences, the proof uses only basic properties of the Jacobi symbol and the law of quadratic reciprocity.

## 1 Jacobi symbols and the norm form $a^2 + 15b^2$

For a prime  $p$  and a non-zero integer  $N$ , we write  $v_p(N)$  for the exponent of  $p$  in the prime factorization of  $N$ . For an integer  $a$  and an odd prime  $p$ , the Legendre symbol  $(a/p)$  is 0 if  $p \mid a$ , 1 if  $a$  is a non-zero square modulo  $p$ , and  $-1$  otherwise. For an odd positive integer  $n$ , the Jacobi symbol is  $(a/n) = \prod_p (a/p)^{v_p(n)}$ , the product over the prime factors  $p$  of  $n$ . It is multiplicative in  $a$  and in  $n$ , it depends only on  $a$  modulo  $n$ , and it is non-zero exactly when  $a$  and  $n$  are coprime. We use the law of quadratic reciprocity for Jacobi symbols together with its supplements [1, Chapter 5, §2]: for odd positive integers  $m$  and  $n$ ,

$$\begin{aligned} \left(\frac{-1}{n}\right) &= (-1)^{(n-1)/2}, & \left(\frac{2}{n}\right) &= (-1)^{(n^2-1)/8}, \\ \left(\frac{m}{n}\right)\left(\frac{n}{m}\right) &= (-1)^{(m-1)(n-1)/4} \quad \text{if } \gcd(m, n) = 1. \end{aligned}$$

**Lemma 2.** *Let  $p$  be an odd prime with  $(-15/p) = -1$ , and let  $a, b \in \mathbb{Z}$ . If  $p \mid a^2 + 15b^2$ , then  $p \mid a$  and  $p \mid b$ .*

*Proof.* Suppose that  $p \nmid b$ . Choose  $c \in \mathbb{Z}$  with  $bc \equiv 1 \pmod{p}$ . Then  $(ac)^2 \equiv -15(bc)^2 \equiv -15 \pmod{p}$ , so  $-15$  is a square modulo  $p$ . As  $(-15/p) \neq 0$ , it is a non-zero square, so  $(-15/p) = 1$ , a contradiction. Hence  $p \mid b$ , and then  $p \mid a^2$ , so  $p \mid a$ .  $\square$

**Lemma 3.** *Let  $a, b \in \mathbb{Z}$  be not both zero. Then  $a^2 + 15b^2 \neq 0$ , and  $v_p(a^2 + 15b^2)$  is even for every odd prime  $p$  with  $(-15/p) = -1$ .*

*Proof.* Clearly  $a^2 + 15b^2 > 0$ . Let  $p$  be an odd prime with  $(-15/p) = -1$ , let  $p^k$  be the largest power of  $p$  dividing both  $a$  and  $b$ , and write  $a = p^k a'$  and  $b = p^k b'$ . Then  $p$  does not divide both  $a'$  and  $b'$ , so  $p \nmid a'^2 + 15b'^2$  by Lemma 2. Since  $a^2 + 15b^2 = p^{2k}(a'^2 + 15b'^2)$ , we get  $v_p(a^2 + 15b^2) = 2k$ .  $\square$

**Lemma 4.** *Let  $n$  and  $n'$  be odd positive integers with  $n' \equiv n \pmod{60}$ . Then  $(-15/n') = (-15/n)$ .*

*Proof.* Write  $-15 = \varepsilon 2^e q$  with  $\varepsilon = -1$ ,  $e = 0$  and  $q = 15$ . By multiplicativity and the reciprocity laws,

$$\left(\frac{-15}{n}\right) = \left(\frac{\varepsilon}{n}\right) \left(\frac{2}{n}\right)^e (-1)^{(n-1)(q-1)/4} \left(\frac{n}{q}\right).$$

Here  $(\varepsilon/n)$  and  $(-1)^{(n-1)(q-1)/4}$  depend only on  $n$  modulo 4,  $(2/n)$  depends only on  $n$  modulo 8, and  $(n/q)$  depends only on  $n$  modulo  $q$ . Since 60 is divisible by  $4q$ , the claim follows.  $\square$

**Lemma 5.** *Let  $n$  be a non-zero integer, and let  $s$ ,  $u$  and  $m$  be positive integers such that every prime factor of  $s$  divides 30, both  $2s$  and  $60s$  divide  $m$ ,  $u$  is odd,  $(-15/u) = -1$ , and  $n \equiv su \pmod{m}$ . Suppose also that  $n > 0$ . Then there is an odd prime  $p$  with  $(-15/p) = -1$  such that  $v_p(n)$  is odd.*

*Proof.* Since  $s \mid m$ , we have  $s \mid n$ , and  $n' = n/s$  satisfies  $n' \equiv u \pmod{m/s}$ . Since  $m/s$  is even and divisible by 60,  $n'$  is odd and  $n' \equiv u \pmod{60}$ . Since  $n > 0$ , Lemma 4 gives  $(-15/n') = (-15/u) = -1$ . By multiplicativity, the product of  $(-15/p)^{v_p(n')}$  over the prime factors  $p$  of  $n'$  is  $-1$ , so some prime factor  $p$  of  $n'$  satisfies  $(-15/p) = -1$  and  $v_p(n')$  is odd. Then  $p$  is odd, and  $p \nmid 30$ , so  $p \nmid s$  and  $v_p(n) = v_p(n')$  is odd.  $\square$

## 2 The auxiliary polynomials

Put  $F(x, y) = 2x^4 + x + y^3 - y^2 - 2$ , so that (1) is the equation  $F(x, y) = 0$ . Define

$$\begin{aligned} H &= 5x^2 - xy - 2x + 2y^2 - y - 1, \\ T &= -128x^5 - 256x^4y + 928x^4 + 160x^3y - 800x^3 - 288x^2y^2 + 64x^2y \\ &\quad - 32x^2 + 128xy^2 - 224xy + 288x + 32y^2 + 256y - 128, \\ U &= 32x^4 - 29x^3 + 44x^2 - 10x - 26, \\ V &= -11x^3 + 4x^2 + 10x - 6, \\ G &= -512x^4 + 608x^3 - 576x^2y - 288x^2 + 256xy - 352x + 64y + 544, \\ A &= -121836x^2 - 112721x + 36025, \\ B &= -354432x^3 - 135596x^2 - 456889x - 531183. \end{aligned}$$

Expanding both sides, one checks the identities

$$HT - U^2 - 15V^2 = GF, \tag{2}$$

$$AU + BV = 2^4 \cdot 43 \cdot 3271, \tag{3}$$

$$104H = 13(x - 4y + 1)^2 + 3(13x - 3)^2 - 144. \tag{4}$$

## 3 Proof of Theorem 1

Suppose that  $(x, y) \in \mathbb{Z}^2$  is a solution of (1). Then  $F(x, y) = 0$ , and (2) gives

$$HT = U^2 + 15V^2, \tag{5}$$

where  $H$ ,  $T$ ,  $U$  and  $V$  now denote the values of these polynomials at  $(x, y)$ .

**Step 1: congruences.** Checking the  $600^2$  residue pairs modulo 600, we find that (1) has exactly 80 solutions modulo 600, and that at each of them  $H$  is congruent modulo 600 to one of the 30 residues  $r$  listed in Table 1. Each  $r$  is listed there as  $r = su$  with  $s$  and  $u$  positive and  $u$  odd, or as  $r$  if  $s = 1$ , and in each case  $(-15/u) = -1$ . Here for  $s = 1$ ,  $2s = 2$  and  $60s = 60$  divide 600, and for  $s = 2$ , every prime factor of  $s$  divides 30, and  $2s = 4$  and  $60s = 120$  divide 600.

**Step 2: signs.** If  $H \leq 0$ , then (4) gives  $13(x - 4y + 1)^2 \leq 144$  and  $3(13x - 3)^2 \leq 144$ , so  $|x - 4y + 1| \leq 3$  and  $|13x - 3| \leq 6$ . Solving these two linear forms for  $x$  and  $y$ , we find that the only integer points satisfying both inequalities are  $(0, 0)$ ,  $(0, 1)$ , and none of them is a solution of (1) with  $H \leq 0$ . Hence  $H > 0$ .

**Step 3: primes  $p$  with  $(-15/p) = -1$ .** Let  $p$  be an odd prime with  $(-15/p) = -1$ . We show that  $v_p(H)$  is even. By (3),  $U$  and  $V$  are not both zero, so (5) and Lemma 3 show that  $v_p(HT) = v_p(U^2 + 15V^2)$  is even. Suppose that  $p$  divides both  $H$  and  $T$ . Then  $p \mid U^2 + 15V^2$  by (5), so  $p \mid U$  and  $p \mid V$  by Lemma 2, and  $p \mid 2^4 \cdot 43 \cdot 3271$  by (3). As  $p$  is odd and  $p \neq 3271$  because  $(-15/3271) = 1$ , we get  $p = 43$ . Checking the  $43^2$  residue pairs modulo 43, we find that  $F$ ,  $H$  and  $T$  have no common zero modulo 43, a contradiction. Hence  $p$  does not divide both  $H$  and  $T$ . If  $p \nmid T$ , then  $v_p(H) = v_p(HT)$  is even, and if  $p \nmid H$ , then  $v_p(H) = 0$ .

**Step 4: contradiction.** By Step 1,  $H \equiv su \pmod{600}$  with  $s$  and  $u$  as in Lemma 5, and  $H > 0$  by Step 2. Hence Lemma 5 applies to  $n = H$ , so there is an odd prime  $p$  with  $(-15/p) = -1$  such that  $v_p(H)$  is odd. This contradicts Step 3, and the theorem is proved.  $\square$

## References

- [1] K. Ireland and M. Rosen, *A Classical Introduction to Modern Number Theory*, 2nd ed., Graduate Texts in Mathematics 84, Springer, New York, 1990.

Table 1: The residues  $r = su$  of  $H$  modulo 600 in Step 1.

|                   |                     |     |                     |                     |                     |
|-------------------|---------------------|-----|---------------------|---------------------|---------------------|
| 11                | 131                 | 221 | $326 = 2 \cdot 163$ | 431                 | 521                 |
| 41                | 149                 | 251 | 341                 | $446 = 2 \cdot 223$ | 551                 |
| 71                | 161                 | 281 | 371                 | 449                 | $566 = 2 \cdot 283$ |
| $86 = 2 \cdot 43$ | 191                 | 299 | $374 = 2 \cdot 187$ | 461                 | 581                 |
| 101               | $206 = 2 \cdot 103$ | 311 | 401                 | 491                 | 599                 |